JF1 1CP Battery Thermal Simulation

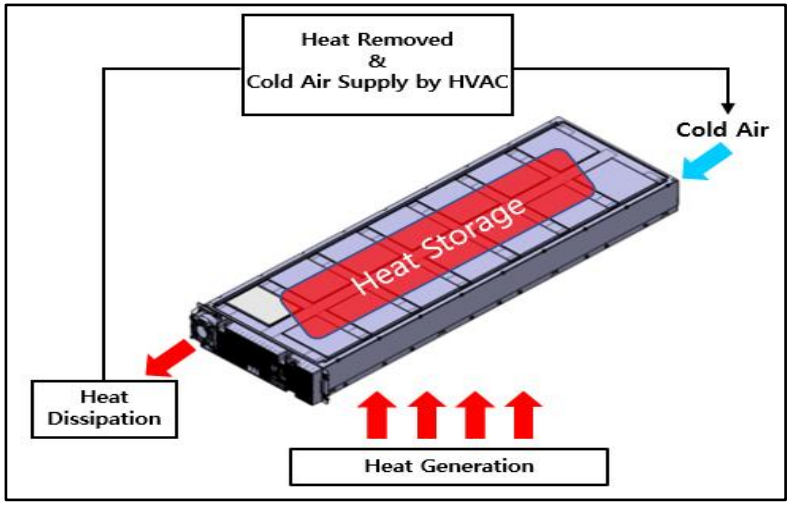
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JF1 1CP Battery Thermal Simulation

- **Governing Equation Based on Energy Conservation Law**

  Battery generates heat when operating during charging/discharging. Based on the cooling performance of LGES's battery Pack, some of heat generation is removed from the battery and the rest of it is stored in the battery, increasing battery temperature.
  This mechanism can be expressed according to energy conservation law as below :



$$\text{Energy(storage)} = \text{Energy(generated)} + \text{Energy(in)} - \text{Energy(out)}$$

Heat Storage | Heat Generation | Heat Dissipation

$$mC_p \frac{dT_{cell}}{dt} = Heat_{gen} - (T_{cell} - T_a)/R_{th}$$

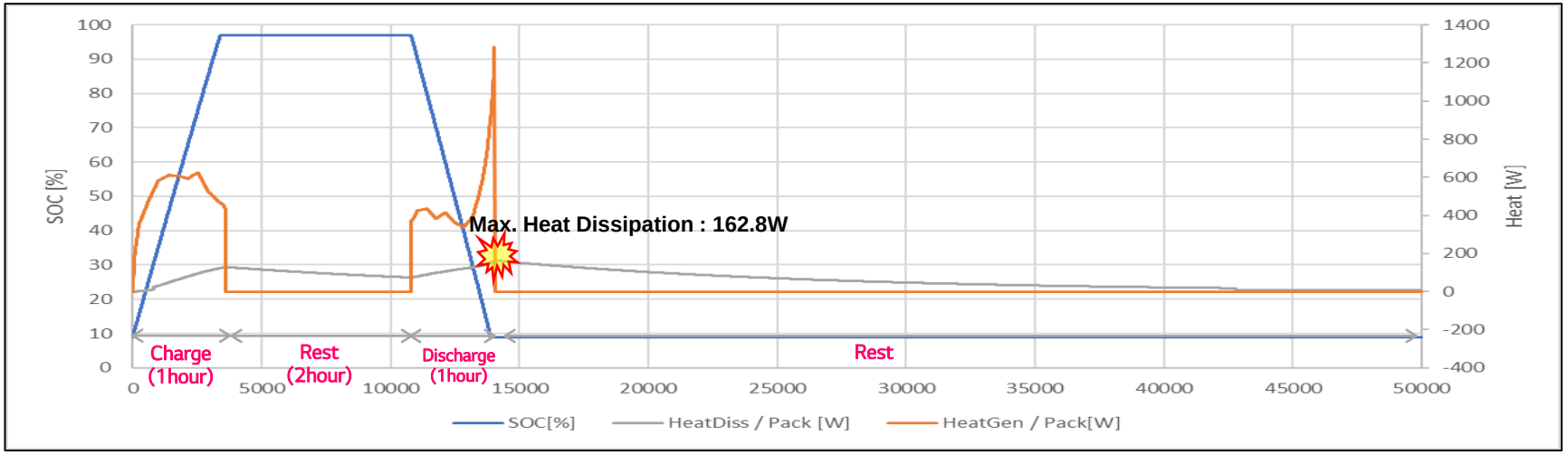
To be removed by HVAC (Required Cooling Capacity)

Rate of Temperature Change | Rate of Internal Heat Generation | Rate of Convection Cooling

R_{th} : thermal resistance coefficient or a reciprocal term of overall heat transfer coefficient(hA) of battery Pack

- **Battery Thermal Simulation : JF1 1CP BOL @ 23degC**

  From battery thermal simulation during 1 cycle, temperature, heat generation and heat dissipation (based on which HVAC capacity is determined) can be obtained.

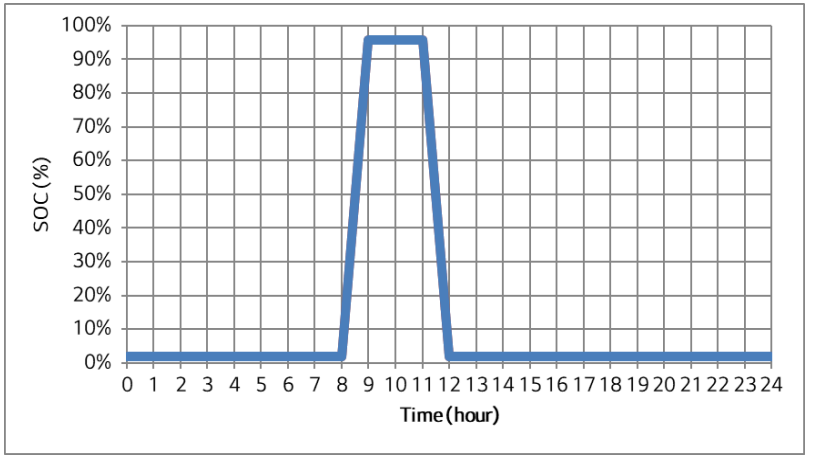


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JF1 1CP Battery Thermal Simulation

❑ Simulation Condition

- Pattern : Peak shifting (DOD Fixed), 2hr Rest
- Num of Cycle : 1cycle / day
- Charge / Discharge CP-rate : 1CP
- DOD : 93.5% (SOC 3.5% ~ 97%)
- Ambient Temp : 23°C
- Voltage Range : 2.7 ~ 3.6V



❑ Simulation Result

- Max Heat Dissipation / Pack : 162.8W (@BOL) / 141.2W (@EOL)
- Pack Fan Heat Generation /Pack : 10.46W
- BPU Fan : BPU heat dissipation is already included in Pack heat dissipation value.

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